AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956

SCHOOL OF ARTIFICIAL INTELLIGENCE COIMBATORE

Causal Weather-to-Humanitarian Impact Forecasting An LLM-Powered Early Warning System

Team AB11

Gowtham S D - CB.AI.U4AID23113
Guhan K B - CB.AI.U4AID23114
Vaishnavi Gupta - CB.AI.U4AID23149
Vyshnav Kumar S - CB.AI.U4AID23151

Mentor : Dr.Unnikrishnan P

Introduction & Motivation

In 2022, East Africa faced a drought, leading to a famine declaration by September, affecting 4.3 million people with acute hunger.

- No integrated system currently connects weather patterns to agriculture and food security - Forecasting is manual, reactive, and isolated.
- Early warning systems could save over 100,000 lives annually and proactive measures are 90% cheaper than emergency responses.
- UN, WFP, and NGOs need predictive systems linking weather data to food security outcomes.

Literature Review

S.No	Paper title and year	Key contributions	Limitations
1	**DisasterNet: Causal Bayesian networks with normalizing flows for cascading hazards (2023)**	Causal Bayesian network for cascading hazards. Normalizing flows + variational inference for dependency modeling.	Post-disaster focus, not pre-event forecasting. Earthquake-centered setting.
2	**Improving disaster resilience with causal machine learning for flood damage estimation (2025)**	CINN for flood damage prediction. DECI for causal structure, plus causal regularization.	Depends on causal discovery quality. Narrow flood - specific scope.
3	**An Explainable Forecasting System for Humanitarian Needs Assessment (2024)**	Foresight for displacement forecasting. Explainable ML pipeline with scenario forecasts.	System paper, not new model design. Context-specific data dependence.
4	**Impact-based humanitarian forecasting using machine learning for floods (2025)**	ML models for flood impact forecasting. Compares models for lead time, scalability, trigger design.	Review paper, not a new model. Focused on floods only.

Literature Review (Cont.)

5	**Using machine learning to forecast conflict events for use in displacement modeling (2025)**	Random Forest conflict forecast. Flee ABM for displacement simulation.	Error propagation across stages. Needs recalibration across regions.
6	**FloodDamageCast: Building flood damage nowcasting with machine learning and data augmentation (2024)**	ML nowcasting for flood damage. Data augmentation for sparse-event learning.	Single-region/event setting. Short horizon, not long-lead warning.
7	**Impact forecasting for humanitarianism: Opportunities and challenges (2025)**	Cross-domain impact forecasting framework. Links weather, conflict, livelihoods, prices.	Conceptual, not algorithmic. No single model proposed.
8	**Machine learning for humanitarian forecasting: A Survey (2025)**	Reviews ML for conflict forecasting. Assesses real-world feasibility.	Survey only. Conflict-focused, not full weather-to-impact pipeline.

Research Gap

- No unified framework combines weather forecasting, causal knowledge, and LLM reasoning for humanitarian early warning.
- Most systems rely on rule-based or statistical methods, limiting their ability to adapt to unseen scenarios.
- Existing solutions provide predictions without explainable causal reasoning or uncertainty estimates.

Problem Statement

- Weather forecasters can predict agricultural shocks months in advance, but no system translates that signal into humanitarian action.
- The causal chain — rainfall anomaly → crop failure → price shock → food insecurity — is assessed manually, after the fact, by siloed experts.
- Meteorologists, agronomists, and humanitarian analysts don't share a common pipeline linking their domains

Objectives

- Develop a causal AI model for weather-to-humanitarian impact forecasting.
- Design an LLM-powered early warning and decision-support system.
- Integrate weather, exposure, vulnerability, and historical data for impact prediction.
- Enable timely and explainable forecasts for proactive humanitarian response.

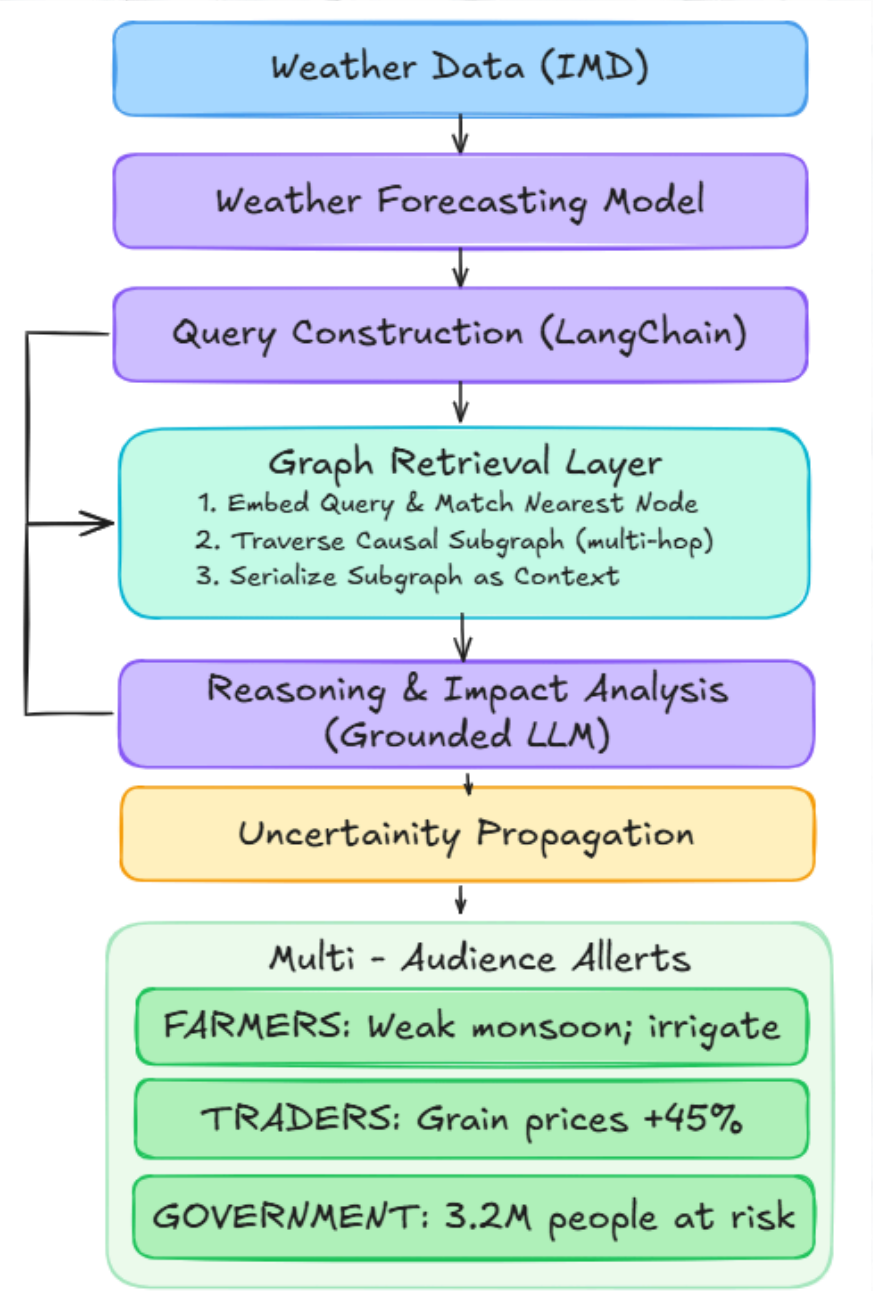
Proposed Methodology

- Forecasting: Ingests IMD weather data via Transformers to generate quantitative deviation triggers.
- Graph Retrieval: LangChain structures queries for Neo4j to traverse multi-hop causal subgraphs.
- Grounded Reasoning: LLMs evaluate impacts using retrieved graph evidence while propagating forecast uncertainties.
- Targeted Outputs: Delivers tailored operational, market, and policy alerts for multiple audiences.

Datasets Required

S.No	Dataset	Source	Role
1	IMD Weather Forecast	India Meteorological Department	Provides rainfall, temperature, humidity, and weather forecasts that trigger the humanitarian impact prediction.
2	ICAR Crop Statistics & Crop Calendar	Indian Council of Agricultural Research (ICAR) / Ministry of Agriculture	Provides district-wise crop information and crop seasons to identify weather-exposed agricultural systems.
3	Sentinel-2 Satellite Imagery (NDVI)	ESA Copernicus	Used to compute NDVI (Normalized Difference Vegetation Index) to monitor vegetation health and validate crop stress caused by weather anomalies.
4	Indian Census	Office of the Registrar General & Census Commissioner	Provides district-wise population and demographic data to estimate human exposure and vulnerability.

Architecture Flow



Forecast-to-Decision Pipeline

- **Detect Weather Anomalies:** Analyze IMD weather forecasts to identify rainfall, temperature, and other climate anomalies that trigger impact assessment.
- **Retrieve Causal Evidence:** Use LangChain and GraphRAG to retrieve relevant causal relationships from the Neo4j knowledge graph.
- **Generate Explainable Predictions:** Grounded LLM analyzes the retrieved evidence to predict and explain potential humanitarian impacts.
- **Quantify Risk & Confidence:** Estimate prediction confidence by propagating uncertainty from weather forecasts and causal reasoning.
- **Deliver Actionable Early Warnings:** Generate stakeholder-specific alerts to support timely humanitarian planning and response.

Weather Forecast → Causal Knowledge Retrieval → Explainable AI Reasoning → Risk Assessment → Early Humanitarian Action

Techniques & Tools

Layer / Component	Techniques & Methodology	Tools & Frameworks
Weather Forecasting	Quantitative deviation triggers & deep learning forecasting	IMD Weather Data, Transformer Models
Query Construction	Natural language to structured retrieval query parsing	LangChain
Graph Retrieval (GraphRAG)	Entity linking, vector similarity matching, multi-hop causal subgraph traversal	Neo4j, Vector Index
Reasoning & Uncertainty	Grounded evidence analysis, confidence interval & R2 error propagation	Grounded LLM, Statistical Propagation Engine

Expected Outcomes

- Product: Deployable API + Dashboard; pilot ready for WFP/UN OCHA
- Social Impact: 8-12 week early warning → pre-positioned aid → famine prevention.
- Metrics: Forecast accuracy vs. IPC, lead time, counterfactual lives saved.
- Scalability: Toolkit for conflict forecasting, disease outbreaks, supply chain disruption.

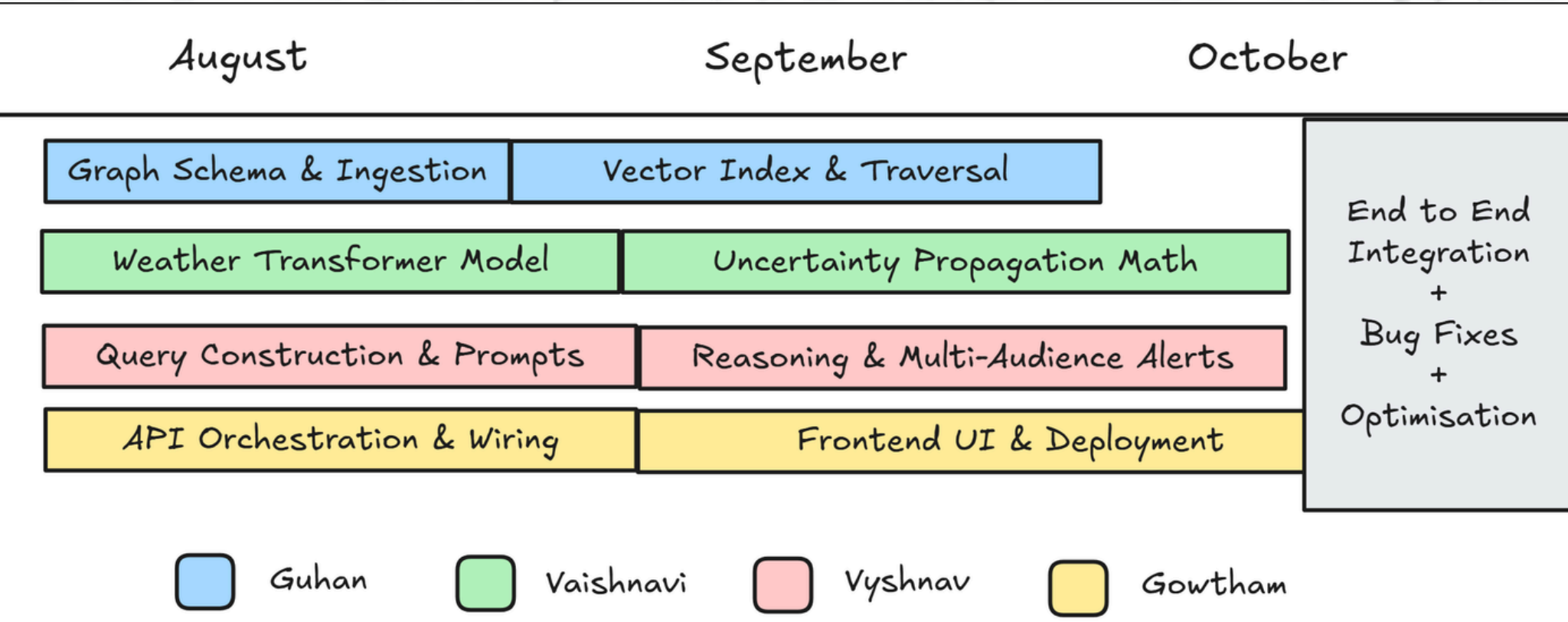
Market Need & Impact

- Global Burden: 258M people in acute hunger (WFP 2024)
- Current Cost: $41B annual humanitarian response spend
- Savings Potential: $30B+ annually + prevention of 100,000+ deaths
- Lead Time Gap: Agencies need 8-12 weeks; systems provide 2-4 weeks
- Preventable Deaths: Early warning could prevent 7000% of deaths at 1/10 cost

Risk Management & Mitigation

- LLM Hallucination → Constrain to JSON; validate against knowledge graph
- Limited Ground Truth → Use historical IPC + price volatility as proxy
- Knowledge Graph Gaps → Start with IPCC; iteratively add from literature
- Computational Cost → Use quantized models; cache queries; batch process
- Data Gaps → Use satellite proxies; prioritize high-risk zones

Timeline & Roles



References

[1] N. Altay and A. Narayanan, “Forecasting in humanitarian operations: Literature review and research needs,” 2022.
[2] A. S. Tozier de la Poterie, W. E. Jjemba, R. Singh, E. Coughlan de Perez, C. V. Costella, and J. Arrighi, “Understanding the use of 2015–2016 El Niño forecasts in shaping early humanitarian action in Eastern and Southern Africa,” 2018.
[3] M. Reichstein et al., “Early warning of complex climate risk with integrated artificial intelligence,” 2025.
[4] K. Baylis and E. C. Lentz, “Impact forecasting for humanitarianism: Opportunities and challenges,” 2025.
[5] P. M. Kam et al., “Impact-based forecasting of tropical cyclone-related human displacement to support anticipatory action,” 2024.
[6] V. L. Boult et al., “Towards drought impact-based forecasting in a multi-hazard context,” 2022.
[7] J. J. L. Westerveld et al., “Forecasting transitions in the state of food security with machine learning using transferable features,” 2021.
[8] G. Martini, A. Bracci, A. Husain, E. Omodei, L. Riches, S. Jaiswal, M. Corea, and J. Rivers, “Machine learning can guide food security efforts when primary data are not available,” 2022.